Karim Wael

Software Engineer | AI Systems

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PROFESSIONAL SUMMARY

Product-focused **Software Engineer** specializing in building scalable backend systems and AI-powered solutions. Proven track record in leading R&D teams, bridging hardware-software gaps, and delivering robust, data-driven solutions in competitive environments like the NASA Space Apps Challenge.

TECHNICAL SKILLS

- **Languages:** C++ (STL, Procedural, OOP), Python (ML, Research), JavaScript (ES6+), TypeScript, PHP (MVC-lite).
- **Web & Systems:** React, Next.js, Node.js, Express, Socket.io, Tailwind CSS, Three.js (WebGL).
- **Tools & Infrastructure:** Docker, AWS, Linux, Git/GitHub, Arduino & Embedded Systems.
- **Databases:** PostgreSQL, MySQL, MongoDB.
- **AI/ML:** Scikit-Learn, Deep Learning, RAG, Computer Vision (HCIA-AI Certified).

PROFESSIONAL EXPERIENCE

IEEE MIU Student Branch | *Cairo, Egypt*

R&D Co-Head | *Sep 2025 – Present*

- Orchestrating technical R&D operations for the student branch, leading a cross-functional team of 30+ students.
- Architecting the **IEEEMIU-PORTAL**, a custom-built PHP MVC management system serving 100+ active users with automated grading, real-time dashboards, and certification workflows.
- Directing student-led hardware-software prototype development for university-wide technical competitions.
- **Arduino & Python Junior Program (Lead Mentor):** Delivered 20+ hours of hands-on technical training.

Technical Committee Member | *Oct 2024 – Oct 2025*

- One-year tenure within the core technical committee, responsible for protocol research and hardware integration.
- Developed hardware-software prototypes for inter-branch robotics competitions and university events.
- Completed intensive engineering curriculum including Python fundamentals and HackerRank certified assessments.

NASA Space Apps Challenge | *Team Moon Walkers* **Technical Team Lead** | *Oct 2024*

- Directed the development of **ProbCast**, an environmental intelligence platform utilizing the NASA POWER API.
- Designed data pipelines to transform raw Earth observation data into actionable reports for event logistics.

MSP Tech Club — MIU | *Cairo, Egypt* **Public Relations Member** | *Nov 2025 – Present*

- Driving technical outreach and club engagement strategies to bridge the gap between engineering concepts and student community awareness.

FEATURED PROJECTS

TALKY | High-Concurrency Real-Time Messaging
C++, Node.js, Socket.io, React, Docker

- Developed a hybrid messaging platform using a **C++ backend** for sub-millisecond message routing with real-time WebSocket state synchronization.

IEEEMIU-PORTAL | Enterprise Branch Management
PHP, MySQL, Vanilla CSS (Glassmorphism), JavaScript

- Engineered a university portal featuring a Learning Management System (LMS) and member dashboards, automating daily operations and serving 100+ active members.

GheitBot | AI-Powered Farming Assistant
Python, Scikit-Learn, Streamlit, OpenWeather API

- Built an AI assistant using a Random Forest model trained on soil and climate data, providing optimal crop rotation schedules and real-time agricultural intelligence.

EDUCATION

Misr International University (MIU) | *Cairo, Egypt* **Bachelor of Science in Computer Science** | *2024 – 2028 (Expected)*

CERTIFICATIONS & AWARDS

- **HCIA-AI V4.0 (Artificial Intelligence)** – Huawei ICT Academy (Feb 2026)
- **Fundamentals of Deep Learning** – Nvidia Deep Learning Institute (Feb 2026)
- **AWS Academy Graduate - AWS Academy Cloud Foundations**
- **AWS Academy Graduate - AWS Academy Machine Learning Foundations**
- **Certificate of Appreciation** – Course Assistant (Arduino/Python Program @ IEEE MIU)